

1.

1.1.

2022 pp. 40 - 41

1.2.

2022 pp. 43, 47 - 54

12 $X\mathrm{cm}^2$ Y g

(1) $Y = aX + b$

(2)

(3)

$$\begin{aligned}(X, Y) = (11, 23), \quad (23, 49), \quad (5, 10), \quad (50, 115), \quad (32, 64), \quad (21, 45)\\ (16, 35), \quad (39, 84), \quad (25, 50), \quad (45, 100), \quad (5, 11), \quad (34, 70)\end{aligned}$$

2.

2.1.

2022 2022 2022 p. 65

(1)

(1)

(2) (2) (2)

(1) (2)

(3) 1 6 (2)

1 2 3 4 5 6 (2) $\Omega = \{1, 2, 3, 4, 5, 6\}$ (3)

(2) Ω (1)

(3) (2)

2.2.

2022 2022 2022 pp. 78, 80, 83

(1)

(2)

(3)

(4) 3

2.3.

2022 2024 2026 p. 78

1 X

c

$P(X = x) = c \times x \quad (x = 1, 2, 3, 4, 5, 6)$ $\times$

(1) c

(2) $x = 1, 2, 3, 4, 5, 6$ $P(X = x)$

(3) X $E[X]$

(4) $P(X \geq 4)$

2.4.

2023 pp. 81, 86

$$f(x) = \begin{cases} \dfrac{c}{x^3} (x \geq 10) \\ 0 \quad (x < 10) \end{cases}$$

(1) c

(2)

(4)

(5)

(6) 1

2.8.

2025

(1) X μ σ^2

(2) 200 60 6 42

78

3.

3.1.

2024 p. 107

6 4 1 2

(1) 2

(2) 2

3.2.

2026 p. 107

3 5

(1) 1 1

(a) 2

(b) 2

(c) 2 1

(2) 2

1

(3) (1) (2)

3.3.

2022 pp. 107 - 108

1 2 3 4 2

X_1 X_2 $Y = |X_1 - X_2|$

(1) Y

(2) Y

(3) Y

3.4.

2023 2025 pp. 113 - 114

(X, Y)

$$f(x,y) = \begin{cases} 6e^{-2x-3y} (x \geq 0, y \geq 0) \\ 0 \qquad\qquad (\text{otherwise}) \end{cases}$$

X Y

3.5.

2026

X Y

$$\int_0^\infty e^{-x}dx = 1, \qquad \int_0^\infty xe^{-x}dx = 1$$

(1) $f(x,y) = \begin{cases} xe^{-x-y} & (x \geq 0, y \geq 0) \\ 0 & (\text{otherwise}) \end{cases}$

(a) $f_X(x)$ $f_Y(y)$

(b) X Y

(2) $f(x,y) = \begin{cases} c(x+y)e^{-x-y} & (x \geq 0, y \geq 0) \\ 0 & (\text{otherwise}) \end{cases}$

(a) c

(b) X Y

3.6.

2022 pp. 103 - 104, 113

1 4 2 2 1 $X = 0$ $X = 1$ 2 $Y = 0$ $Y = 1$

(1) (X, Y)

(2) X Y

3.7.

2022 pp. 67, 114 - 115

2 A B

3.8.

2022 2023 2026 p. 115

3 $\frac{1}{2}$

(1) 1 2

(2) 1 2 A 2 3
B A B

(3) $P(A|B)$

3.9.

2024 p. 115

2 A 4 B 2
7 C 2 9

(1) A B

(2) A C

3.10.

2023 pp. 72 - 73, 109 - 110, 114 - 115

2 E F $P(E) = a$ $P(F) = b$

(1) $P(E \cup F)$ a b

(2) $P(E \cap F^c | E \cup F)$ a b F^c F

3.11.

2025

1 5
X Y $Z = X - Y$

(1) X Y

(2) Z

(3) Z

(4) Z

3.12.

2025 pp. 118 - 119

(1) 2 A B

(2) 500 1 X X
2 K_1 K_2 3 : 1
K_1,K_2 0.98 0.90
0.01
K_1

3.13.

2022 2023 pp. 118 - 119

3 5 1
2

(1) 1

(2) 2 1

3.14.

2023 pp. 118 - 119

1 10% 3

(1)

(2) 2

3.15.

2022 pp. 118 - 119

5 4 3
1 100 g 45 g 30 g

(1) 1
40 g

(2) 2
100 g 2

(3) 3
150 g 3

3.16.

2022 2024 pp. 118 - 119

A 0.1%
A /
99% 10%

(1)

(2) A 1

3.17.

2023 pp. 118 - 119

99%
99% 0.5%
1

(1)

(2)

(3)

(4)

3.18.

2023 pp. 118 - 119

S S
K

- S K 99% 1%
- S K 99% 1%

(1) K S

(2) (1)

3.19.

2025 pp. 118 - 119

A

B

(1) $P(A|B) = 0.98$ $P(A|B^c) = 0.005$ $P(B) = 0.001$

(2) $P(A|B^c) = 0.003$ $P(B) = 0.0005$ $P(B|A) \geq 0.142$ $P(A|B)$

3.20.

pp. 118 - 119

500 1 XXX 2

K_1 K_2 $K_1 : K_2 = 3 : 1$

$\boldsymbol{K_1}$ 0.98

$\boldsymbol{K_2}$ 0.90

0.01

(1) K_1

(2) (1)

3.21.

2023 pp. 122 - 124

p $0 < p < 1$ X Y

$$f(k) = \begin{cases} p & (k = 1) \\ 1 - p & (k = -1) \end{cases}$$

$Z = XY$

(1) $E[Z]$

(2) X Z $E[(X - E[X])(Z - E[Z])]$

(3) X Z p p Y Z

(4) (3) p $\Pr[X + Y + Z \leq 2]$

3.22.

2023 2024 pp. 125 - 126

2 X Y

(1) a

(2) X Y $\mathrm{Cov}(X, Y)$

(3) X Y

	$Y = 1$	$Y = 10$	$Y = 100$
$X = 1$	a	a	a
$X = 2$	0	a	a
$X = 3$	a	a	a

3.23. n

2023 p. 127

n 1 n $(n + 1)$

4.

4.1.

2025

2 $B(2, p)$

4.2.

2024 p. 141

1 6

(1) 10 5

(2) 100 25

4.3.

2023 2026 pp. 93 - 95, 141

p n

X 2 $\mathrm{Bin}(n, p)$ $q = \left(\dfrac{X}{n}\right)$

(1) $P(|q - p| \leq 0.02) \geq 0.95$ n

(2) (1) n (2)

(3) p

(3) (1) -

n

(4) (2) (3) n

4.4.

2024 pp. 143 - 145, 153

(1) :

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (-\infty < x < \infty)$$

$y = f(x)$ (x, y)

(2) /

(3)

4.5.

2022 pp. 121 - 122, 143

(1) X Y $N(170, 6^2)$

(a) $W = \dfrac{X+Y}{2}$

(b) X Y W

(2) X Y

$$f(x) = \frac{1}{2\pi} \quad (0 \leq x \leq 2\pi), \qquad g(x) = \frac{1}{12} \quad (-3 \leq x \leq 9)$$

$W = XY$

4.6.

2024 pp. 147 - 151

(1) X $N(\mu, \sigma^2)$

(a) $P(\mu - \sigma \leq X \leq \mu + \sigma)$

(b) $P(\mu - 2\sigma \leq X \leq \mu + 2\sigma)$

(c) $P(\mu - 3\sigma \leq X \leq \mu + 3\sigma)$

(2) (

)

(3)

(a) $z(0.01)$

(b) $\Phi(z_0) = 0.01$ z_0

(c) $z(0.005)$

4.7.

2024 pp. 149 - 151

(1) a Z $N(0, 1)$

$$P(|Z| \geq a) = 2P(Z \geq a)$$

(2) X $N(m, \sigma^2)$

$$P\left(|X - m| \geq \frac{\sigma}{4}\right)$$

4

(3) m σ n

$\overline{X}$

$$P\left(|\overline{X} - m| \geq \frac{\sigma}{4}\right) \leq 0.02$$

n

4.8.

2024 2026 pp. 149 - 150

4

E 350,198 66.31

23.55 X $N(66.31, (23.55)^2)$

(1) $P(X \geq 80)$

(2) (3-a)

(3) 80 95

(4) 20%

4.9.

2025 pp. 149 - 150

4

120

10 100 15

C

(1) 95% C

(2)

4.10.

2022 pp. 153 - 154

(1) 6

(2) 10000

(3) 10000 4900 5100

4.11.

pp. 93 - 95, 153 - 154

200 60 6

X

(1) $42 \leq X \leq 78$

(2)

(3) X $N(60, 6^2)$ $P(42 \leq X \leq 78)$

5.

5.1.

2023 2025 pp. 168 - 169

2

5.2.

2024 pp. 93, 168 - 169

1 ATM ATM 1 40

ATM 1 60

ATM

/

60

75

90

105

120

5.3.

2023 pp. 166 - 169

(1) 3 5

(2) 1 2 1 4

5.4.

2025

4

(1) 1 3

2

(2) X $P(X=3)=5P(X=5)$

$P(X \leq 3)$

5.5.

2023 pp. 166 - 169

1% 200

(1) X

(2)

(3) 3

(2) (3)

5.6.

2024 pp. 153 - 154, 166 - 169

4

(1) 240 6 38 45

(2) 0.1% 500 3

5.7.

2026

$p=0.001$ $n=500$

X

$$e^{-0.5} \approx 0.6065, \qquad \sqrt{0.5} \approx 0.7071$$

(1) (a) X

(b) $E[X]$ $\mathrm{Var}(X)$

(2)

(a)

(b) $P(X \geq 3)$

(3)

(a)

(b) $P(X \geq 3)$

(4)

(a) (2) (3)

(b)

np

5.8.

2022 2023 pp. 139 - 143, 168 - 169

p $0 < p < 1$ $(1-p)$ N $N > 1$

(1) $N = 5$

(2) t $0 \leq t \leq N$

(3) m $m =$ N $0 \leq m \leq N$ m

(4) n $0 \leq n \leq N$ $N - n$ $P(n\&N, p)$

(5) (4) n

(6) n $\lambda = Np$ $N \to \infty$ (4) $P(n\&N, p)$ $P(n; \lambda)$ $\lim_{N\to\infty}\left(1 + \frac{x}{N}\right)^N = e^x$

6.

6.1.

2022 p. 174

2

(1)

(2)

6.2.

2025

μ σ^2 n $X_1, X_2, ..., X_n$.

(1) $X_1, X_2, ..., X_n$

(2) .

(3) n $S_n = X_1 + X_2 + \cdots + X_n$

.

(4)

6.3.

2025 p. 179

1180 20

(1) 25 1170

(2) N 0.9 1175

N

6.4.

2025

1.2 kg 0.1 kg

100 122.5 kg

6.5.

2022 pp. 202 - 205

μ σ^2 3 X_1 X_2 X_3

(1) .

(a) $T_1 = X_1 + X_2 - X_3$

(b) $T_2 = \dfrac{X_1 + 2X_2 + X_3}{4}$

(c) $T_3 = \dfrac{X_1 + X_2 + X_3}{3}$

μ

(2) $\{T_1, T_2, T_3\}$

6.6.

2023 2023 2025 pp. 176 - 180, 202 - 205

(1) 0 1.2 3

$\{-1, 0, 1\}$

(a)

(b)

(c)

(2)

(3)

(4) θ $\Theta_n = T(X_1, X_2, ..., X_n)$ 3

(a)

(b)

(c)

(5) 3

6.7.

2022 2024 p. 205

1000 10

$$32, \quad 86, \quad 78, \quad 49, \quad 48, \quad 77, \quad 97, \quad 49, \quad 51, \quad 74$$

1000

6.8.

2025 pp. 179, 216

p n

X

(1) X .

(2) n X

(3) p_0 95%

n

(4) $p = 0.05$ $n = 1900$ $P(76 \leq X \leq 114)$

6.9.

2022 pp. 180 - 182

p p 10,000

3,500 $p = \dfrac{3500}{10000} = 0.35$

(1) p

(2) p 2

6.10.

2025

(1) n A Y

$q = \dfrac{Y}{n}$ A p $100(1-\alpha)\%$

(2) S TV T

140 32 T

(a) p 95%

(b) p 3% 0.95

(c)

6.11.

2024 2024 pp. 223 - 225

(1) $Bin(1, p)$ $X_1, X_2, ..., X_n$

(a)

(b)

(2) n A Y

(a) $q = \dfrac{Y}{n}$ A p $100(1-\alpha)\%$

(b)

(3) S TV T

140 32 T

(a) p 95%

(b) p 3% 0.95

(c)

6.12.

2024 pp. 223 - 225

(1) p 0.8 0.9 0.99

$\hat{p}$ p 0.02

(2) p 0.95 $\hat{p}$

p 0.03

6.13.

2025

2 p S $1-p$ F

m S k $B(m;p)$

(1) $B(m;p)$ $f(k)$.

(2) X $B(m;p)$ X .

(3) p $l(p)$.

(4) p .

(5) . .

(a) (4) ?

(b) (4) ?

(c) (4) ?

6.14.

2025

$N(\mu,\sigma^2)$ μ σ^2

(1) μ

(2) σ^2

6.15.

2025

(1)

(2)

/ σ^2 n

N

(3) /

7.

① 仮説の検定

② 統計量の検定

③ 優位水準と棄却域の設定

④ 帰無仮説の棄却・選択

7.1.

2022 2023 pp. 227 - 231

(1)

(2)

(3) (4)

5% (3)

5% (5)

(1)

(1)

(6) (6) (7)

- (8:)
- (9:)

7.2.

2022 pp. 227 - 234

(1)

(2)

7.3.

2023 pp. 2235 - 236

8000

25 7700

5%

640

7.4.

2023 pp. 238 - 240

T

S 365 6

330, 350, 385, 321, 340, 365

5%

7.5.

2025 pp. 238 - 240

T

S 360 6

350, 355, 385, 390, 370, 365

(1) 10%

(2) 5%

7.6.

2023 2024 2024 2025 pp. 235 - 236

(1) μ $N(\mu, 49)$ 49

$H_0 : \mu = 4,$ $H_1 : \mu > 4$

m

(a) 0.05 $m = 6$ 2

(b) 0.01 $m = 7$ 2

(2) $H_1 : \mu = 5$ 0.05

2 0.2 0.1

7.7.

2022 pp. 241 - 242

(1) 2 A B 2

10 12

A 2.04 0.03 B

2.02 0.02 2

5%

2

(2) A B

5%

7.8.

2022 pp. 241 - 242

100 70

68 15

5%

7.9.

2024 pp. 252 - 253

3 1

(1) A 10 A

3 1 A

5%

(2) 10 7 3 1

7.10.

2024 pp. 252 - 253

5 1 5%

7.11.

2023 pp. 252 - 253

10 X p

$H_0 : p = \dfrac{1}{2}$ H_0

(1) X

(2) X

(a) $H_1 : p < \dfrac{1}{2}$ 5%

(b) $H_1 : p \neq \dfrac{1}{2}$ 2%

7.12.

2023 pp. 252 - 253

5

1 p

(1) 5 5 0

(a) $p = \dfrac{1}{2}$

(b) $p = \dfrac{2}{3}$

(2) $p \geq \dfrac{1}{2}$

(a) 3

$p = \dfrac{1}{2}$

(b) $p = \dfrac{3}{4}$ 3

7.13. 2

256 116 5%

.

(1) 2

(2) 2 10%

7.14.

2023 2025 pp. 252 - 253

p $0 < p < 1$

$H_0 : p = \dfrac{1}{2}$ 3 3

3 H_0 H_0

(1) 1 1

(2) 2 2

(3) 2 $\dfrac{1}{3}$ p

7.15.

2024 pp. 252 - 253

3 A B C A 3 B 2 C 1 1

1

(1) A A $\dfrac{1}{16}$ B B $\dfrac{1}{9}$ C C $\dfrac{1}{25}$

1

(2) 960 640

(1) 5%